

Addressing Modes Summary

Addressing Modes Summary

- Operands can be

- Immediate

e.g. `ADD AX, 5`

- Register

`ADD AX, BX`

- Memory

Addressing Modes Summary (contd.)

- Direct Addressing
 - `mov ax, [num1]`
- Based Register Indirect
 - `Mov [bp], ax`

Addressing Modes Summary (contd.)

- Indexed Register Indirect
 - Mov [si], ax
 - Mov ax, [di]
- Based Register Indirect + Offset
 - Mov [bx+300], ax
 - Mov ax, [bp+300]

Addressing Modes Summary (contd.)

- Indexed Register Indirect + Offset

- `Mov [si+300], ax`

- `Mov ax, [di+300]`

- `Mov [si+num1], ax`

- Base + Index

- `Mov [bx+si], ax`

Addressing Modes Summary (contd.)

- Base + Index + Offset
 - Mov [bx+si+300] , ax
 - Mov [bx+si+num1] , ax

Addressing Modes

- `Mov ax,[num1]`
- `Mov ax,[bx]`
- `Mov ax,[si]`
- `Mov ax,[bx+num1]`
- `Mov ax,[si+num1]`
- `Mov ax,[bx+si]`
- `Mov ax,[bx+si+num1]`

Offset Addressing

- Default Segment=ds
- Mov ax,[0x1234]
- Mov ah,[0x1234]
- Mov byte[0x1234],10
- Mov word[0x1234],10

Base Addressing

- Default Segment=ds
- Mov ax,[bx]
- Mov byte[bx],10
- Mov word[bx],10
- Default segment =ss
- Mov ax,[bp]
- Mov byte[bp],10
- Mov word[bp],10

Base+Offset addressing

- Default segment=ss
 - Mov byte[bp+0x10]=10
 - Mov word[bp+0x10]=10
- Default segment=ds
 - Mov ax,[bx+0x100]
 - Mov byte[bx+0x100],10
 - Mov word[bx+0x100],10

Index+offset addressing

- Default segment=ds
 - `Mov ax,[si+0x0100]`
 - `Mov ax,[di+0x0100]`
 - `Mov byte[si+0x0100],10`
 - `Mov word[si+0x0100],10`
 - `Mov byte[di+0x100],10`
 - `Mov word[di+0x100],10`

Index addressing

- Default segment=ds
 - `Mov ax,[si]`
 - `Mov ax,[di]`
 - `Mov byte[si],10`
 - `Mov word[di],10`

Base+index addressing

- Default segment=ds
 - `Mov ax,[bx+si]`
 - `Mov byte[bx+si],10`
 - `Move word[bx+di],10`
- Default segment=ss
 - `Mov word[bp+si],10`
 - `Mov byte[bp+di],10`

Base +Index +Offset addressing

- Default segment=ds
 - `Mov ax,[bx+si+0x100]`
 - `Mov byte[bx+si+0x100],10`
 - `Mov word[bx+si+0x100],10`
- Default segment=ss
 - `Mov byte[bp+si+0x100],10`
 - `Mov word[bp+si+0x100],10`

Common Mistakes

- Address is 16-bit long, contained in 16-bit Register
- Mov al,[bl] **Not** allowed
- Mov ax,[bx-si]
- [BX+SI] is allowed, [BX-SI] is **NOT** allowed
- [Base+Base], [Index+Index] both **NOT** allowed,